

Computability: Exercise 8

Due: June 1, 2026

Question 1

Recall that a *Hamiltonian path* in a graph is a path that goes through every vertex exactly once. A *Hamiltonian cycle* is a path that goes through every vertex exactly once, except one vertex from which the path starts and in which it ends.

Some related decision problems are:

$\text{D-HAMPATH} \stackrel{\text{def}}{=} \{\langle G \rangle : G \text{ is a directed graph with a Hamiltonian path}\}$

$\text{D-ST-HAMPATH} \stackrel{\text{def}}{=} \{\langle G, s, t \rangle : G \text{ is a directed graph with a Hamiltonian path from } s \text{ to } t\}$

$\text{D-HAMCYCLE} \stackrel{\text{def}}{=} \{\langle G \rangle : G \text{ is a directed graph with a Hamiltonian cycle}\}$

For undirected graphs we similarly define U-HAMPATH, U-ST-HAMPATH and U-HAMCYCLE. Prove the following.

- a. $\text{D-ST-HAMPATH} \leq_p \text{D-HAMPATH}$

Hint: Consider a small addition to the graph.

- b. $\text{D-ST-HAMPATH} \leq_p \text{D-HAMCYCLE}$

Hint: Consider a small addition to the graph.

Question 2

Definition 1. For a given set $U = \{1, \dots, n\}$ and subsets $S_1, \dots, S_m \subseteq U$, a *set cover* is a collection of subsets $S_{i_1}, \dots, S_{i_\ell}$ such that their union is U . The *size of the set cover* is the number of subsets in the collection. The corresponding decision problem:

$\text{SET-COVER} \stackrel{\text{def}}{=} \{\langle U, S_1, \dots, S_m, k \rangle : \text{There is a set cover of size } k\}.$

Prove that SET-COVER is NP-complete. For NP-hardness, use a reduction from VC (namely the vertex cover problem).

Question 3

Definition 2. Let $G = (V, E)$ be a graph. A *dominating set* is a set of vertices $S \subseteq V$ such that for every $v \in V$ we have: $v \in S$ or v has a neighbor $u \in S$. The corresponding decision problem:

$\text{DS} \stackrel{\text{def}}{=} \{\langle G, k \rangle : G \text{ has a dominating set of size } k\}.$

Prove that DS is NP-complete.

Hint: For NP-hardness, use a reduction from VC, and add something to each edge.

Question 4

The travelling salesman problem, in its decision problem version, is:

$$\text{TSP} \stackrel{\text{def}}{=} \{ \langle G, b \rangle : G \text{ is a complete undirected positive-weighted graph} \\ \text{with a Hamiltonian cycle of weight } \leq b \},$$

where a positive-weighted graph is a 3-tuple $G = (V, E, w)$ where V is the set of vertices, $E \subseteq V \times V$ is the set of edges, and $w : E \rightarrow \mathbb{R}_{\geq 0}$ is the weight function that assigns a non-negative weight to each edge. We define the weight of a path as the sum of the costs between following vertices in the path.

Prove that TSP is NP-complete.

Hint: Use the fact that

$$\text{HAM-PATH} \stackrel{\text{def}}{=} \{ \langle G \rangle : G \text{ is an undirected graph with a Hamiltonian path} \}$$

is NP-HARD.